

Exercise 3

AIM :

To create two variables InputValue and OutputValue, get the value of InputValue from user, Now by using Macro READBIT read last 4 bits individually and use SETBIT or RESETBIT macro to change the MSB 4bits of the OutputValue and print the output.

ALGORITHM :

Step 1: Initialize Variables

- InputValue : To store user input (0-255 range)
- OutputValue : To store processed result (initially 0)

Step 2: Define Bit Manipulation Macros

- READBIT(num,bit) : Extracts a specific bit from a number
- SETBIT(num,bit) : Sets a specific bit to 1
- RESETBIT(num,bit) : Sets a specific bit to 0

Step 3: Input Phase

- Prompt user to enter a number
- Read and store in InputValue

Step 4: Processing Phase

- Loop through bits 0 to 3 (last 4 bits) of InputValue
- For each bit position i ;
 - Use READBIT to check if bit i is 1 or 0
 - If bit is 1: Use SETBIT to set bit (4+i) in OutputValue
 - If bit is 0: Use RESETBIT to clear bit (4+i) in OutputValue

Step 5: Output Phase

- Display the final value of OutputValue

OUTPUT :

```
D:\C programming\DHAVAMANI_EX3>gcc main.c -o optr.exe  
  
D:\C programming\DHAVAMANI_EX3>optr.exe  
Enter a number: 1  
  
Reading last 4 bits of InputValue:  
  
OutputValue: 16
```